

Francis Banville

COMPUTATIONAL ECOLOGIST AND DATA SCIENTIST WITH EXPERIENCE IN ONE HEALTH

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Profile

Research interests	computational ecology, biodiversity indicators, biostatistics, ecological networks, One Health
Programming	R, Julia, Git, Bash, Markdown, LaTeX
Specialized software	OpenRefine, QGIS, ArcGIS, Excel
Languages	French (native), English (fluent), Spanish (B2.2)

Education

IN PROGRESS

Microprogram in Higher Education

Université de Montréal

Montreal, CAN

Jan. 2025 - Dec. 2026

- GPA: 4.225 (12 / 15 credits)

COMPLETED

Ph.D. in Biological Sciences

Université de Montréal

Montreal, CAN

Sept. 2019 - Dec. 2024

- option General Biology
- advisors: Drs. Timothée Poisot and Dominique Gravel (Université de Sherbrooke)
- thesis: Towards a maximum entropy theory of food webs (classified as excellent, 84 research credits)
- GPA: 4.150/4.3 (6 course credits)

B.Sc. in Biological Sciences

Université de Montréal

Montreal, CAN

Jan. 2016 - Apr. 2018

- option Biodiversity, Ecology, and Evolution
- GPA: 4.198 / 4.3 (91 credits)

INTERRUPTED

M.Sc. in Biological Sciences

Université de Montréal

Montreal, CAN

Sept. 2018 - Aug. 2019

- option Quantitative and Computational Biology (course-based M.Sc.)
- GPA: 4.217 / 4.3 (35 / 45 credits)

B.Sc. in Mathematics

Université de Montréal

Montreal, CAN

Sept. 2013 - Aug. 2015

- option Applied and Pure Mathematics
- GPA: 4.161 / 4.3 (33 / 90 credits)

Main professional experience

Postdoctoral fellow

Université de Montréal

Montreal, CAN

Jan. 2025 - May 2026

- assessing the usability of the KM-GBF indicators for monitoring One Health actions
- advisors: Drs. Timothée Poisot, Andrea Paz Velez, and Colin Carlson (Yale University)

Consultant

Secretariat of the Convention on Biological Diversity (SCBD)

Montreal, CAN

Jan. 2025 - May 2026

- developing indicators, metrics and progress measurements tools on biodiversity and health
- in collaboration with: the GEO BON One Health working group

Teaching assistant

Université de Montréal, département de sciences biologiques

Montreal, CAN

Sept. 2020 - Dec. 2024

- preparing course material, animating hands-on sessions, grading
- courses: biostatistics, population dynamics, research methods in biology, network theory

Grants and awards

GRANTS

2026	Mobility grant , Les Offices jeunesse internationaux du Québec (LOJIQ)	\$850
2024	Scholarship for end of PhD studies (5th year) , Graduate and Postdoctoral Studies, U. de Montréal	\$8,000
2023-24	Fully funded fellowship , Computational Biodiversity Science & Services program (BIOS ²)	\$10,500
2019-24	Mobility grants , Computational Biodiversity Science & Services program (BIOS ²)	\$10,000
2023	Excellence Scholarship , Fonds de bourses en sciences biologiques, U. de Montréal	\$1,500
2023	Perseverance Scholarship , Département de sciences biologiques, U. de Montréal	\$2,000
2019-23	PhD Excellence Scholarship , Institute for Data Valorization (IVADO)	\$100,000
2019-21	Fast-track Master's to PhD program , Graduate and Postdoctoral Studies, U. de Montréal	\$14,000
2019	Research internship grant , The Lapierre lab, Université de Montréal	\$6,000
2018	Research internship grant , The Lapierre lab, Université de Montréal	\$8,400
2017	Undergraduate Student Research Awards , Natural Sci. and Engr. Research Council of CAN (NSERC)	\$7,625

AWARDS

2024	Best presentation by a PhD student , 34th Biology Symposium of the University of Montreal	U. de Montréal
2014-18	Dean's Award of the Faculty of Arts and Sciences , Distinction for academic excellence	U. de Montréal
2011	Governor General's Academic Medal , Distinction for the best overall average of the 2011 cohort	Polybel high school
2011	Lieutenant Governor's Youth Medal , Honorary medal for academic excellence and social involvement	Polybel high school

Scientific publications

PEER-REVIEWED ARTICLES

[13] Soucy, C.P., Banville, F. , Beasley, E., Lefebvre, B., Gutiérrez-Marco, J.C., Poisot, T., and Cameron, C.B. (2026). A paleoecological trophic network model of the Early Ordovician Lower Fezouata Shale (Morocco) fossil fauna. iScience.	2026
https://doi.org/10.1016/j.isci.2026.116819	
[12] Bolduc, D., Dulude-de Broin, F., Bergeron, G., Villeneuve, C., Weiss-Blais, M., Couloigner, C., Dubourg, R., Fraser-Franco, M., Banville, F. , Moisan, L., Montiglio, P.-O., Fortin, D., Thibault, F., Allard, A., Gaudreau-Rousseau, C., Roca, I., Durand, A., Massé, C., Barreau, E., ... Legagneux, P. (2025). Impersonating predators and prey to study trophic interactions through real-life simulations. Methods in Ecology and Evolution. http://doi.org/10.1111/2041-210x.70180	2025
[11] Banville, F. , Strydom, T., Blyth, P., Brimacombe, C., Catchen, M.D., Dansereau, G., Higino, G., Malpas, T., Mayall, H., Norman, K., Gravel, D., and Poisot, T. (2025). Deciphering probabilistic species interaction networks. Ecology Letters, 28(6), e70161. https://doi.org/10.1111/ele.70161	2025
[10] Strydom, T., Bouskila, S., Banville, F. , Barros, C., Caron, D., Farrell, M.J., Fortin, M.-J., Mercier, B., Pollock, L.J., Runghen, R., Dalla Riva, G.V., and Poisot, T. (2023). Graph embedding and transfer learning can help predict potential species interaction networks despite data limitations. Methods in Ecology and Evolution, 14(12). https://doi.org/10.1111/2041-210X.14228	2023
[9] Banville, F. , Gravel, D., and Poisot, T. (2023). What constrains food webs? A maximum entropy framework for predicting their structure with minimal biases. PLOS Computational Biology, 19(9), e1011458. https://doi.org/10.1371/journal.pcbi.1011458	2023
[8] Higino, G.T., Banville, F. , Dansereau, G., Forero-Muñoz, N.R., Windsor, F., and Poisot, T. (2023). Mismatch between IUCN range maps and species interactions data illustrated using the Serengeti food web. PeerJ, 11, e14620. https://doi.org/10.7717/peerj.14620	2023
[7] Lawlor, J., Banville, F. , Forero-Muñoz, N.R., Hébert, K., Martínez-Lanfranco, J.A., Rogy, P., and MacDonald, A.A.M. (2022). Ten simple rules for teaching yourself R. PLOS Computational Biology, 18(9), e1010372. https://doi.org/10.1371/journal.pcbi.1010372	2022

- [6] Strydom, T., Bouskila, S., **Banville, F.**, Barros, C., Caron, D., Farrell, M.J., Fortin, M.-J., Hemming, V., Mercier, B., Pollock, L.J., Runghen, R., Dalla Riva, G.V., and Poisot, T. (2022). Food web reconstruction through phylogenetic transfer of low-rank network representation. *Methods in Ecology and Evolution*, 13(12), 2838-2849. 2022
<https://doi.org/10.1111/2041-210X.13835>
- [5] Strydom, T., Catchen, M.D., **Banville, F.**, Caron, D., Dansereau, G., Desjardins-Proulx, P., Forero-Muñoz, N.R., Higino, G., Mercier, B., Gonzalez, A., Gravel, D., Pollock, L., and Poisot, T. (2021). A roadmap towards predicting species interaction networks (across space and time). *Philosophical Transactions of the Royal Society B: Biological Sciences*, 376(1837), 20210063. 2021
<https://doi.org/10.1098/rstb.2021.0063>
- [4] Higino G., Forero-Muñoz, N.R., **Banville, F.**, Dansereau, G., and Poisot, T. (2021). Computers can help us find raccoons and other living creatures. *Front. Young Minds*. 9(595275). 2021
<https://doi.org/10.3389/frym.2021.595275>
- [3] **Banville, F.**, Vissault, S., and Poisot, T. (2021). Mangal.jl and EcologicalNetworks.jl: Two complementary packages for analyzing ecological networks in Julia. *Journal of Open Source Software*, 6(61), 2721. 2021
<https://doi.org/10.21105/joss.02721>
- [2] Dansereau, G., **Banville, F.**, Basque, E., MacDonald, A.A.M., and Poisot, T. (2020). [Re] Chaos in a three-species food chain. *ReScience C*, 6(3), 5. 2020
<https://doi.org/10.5281/zenodo.4022518>
- [1] MacDonald, A.A.M., **Banville, F.**, and Poisot, T. (2020). Revisiting the links-species scaling relationship in food webs. *Patterns*, 0(0). 2020
<https://doi.org/10.1016/j.patter.2020.100079>

PREPRINTS

- [3] **Banville, F.**, Burnel, C., Carlson, C.J., Eiffener, E., Munoz Acevedo, G., Paz, A., and Poisot, T. (2026). The Global Biodiversity Framework supports global assessment of One Health actions. *EcoEvoRxiv*. 2026
<https://ecoevorxiv.org/repository/view/13028/>
- [2] Catchen, M.D., **Banville, F.**, Boutin, A.C., Brookson, C.B., Carlson, C.J., Dansereau, G., Gibb, R., Houle, M., Kaza, B., Robertson, H., Simons, D., Seifert, S., and Poisot, T. (2025). A protocol for biodiversity-informed wildlife disease surveillance. *EcoEvoRxiv*. 2025
<https://ecoevorxiv.org/repository/view/11160/>
- [1] Fuster-Calvo, A., Higino, G.T., Parent, C., Caron, D., **Banville, F.**, Massol, F., Blanchet, F.G., Hebert, K., Pollock, L., Maiorano, L., Guimaraes, P.R., Silva, P., Thuiller, W., and Gravel, D. (2025). Tracking community change via network coherence. *bioRxiv*. 2025
<https://doi.org/10.1101/2025.11.04.686602>

PHD THESIS

- Banville, F.** (2024). Vers une théorie de l'entropie maximale des réseaux trophiques. Papyrus, Université de Montréal. 2024
<https://hdl.handle.net/1866/40637>

Other publications

JOURNAL ARTICLES

- [1] Boiteux, M., Di Bacco, K., **Banville, F.**, and Fortin, P. (2026). Les jeunes sont prêts pour l'approche Une seule santé. Qu'en est-il du Québec? *Le Devoir*, Opinion, Idées. 2026
<https://www.ledevoir.com/opinion/idees/979739/jeunes-sont-prets-approche-seule-sante-est-il-quebec>

Presentations

ORGANIZED SESSIONS

[4] Groom, Q., Schwantes, C., **Banville, F.**, Hoyos, J., Marceló-Díaz, D., Upham, N., and Poisot, T. (2025, October 23). Biodiversity data for One Health: Integrating open data to address global health challenges [Symposium]. Living Data 2025, Bogota, Colombia. 2025

[3] Arce Plata, M.I., **Banville, F.**, Cruz Rodríguez, C., Dansereau, G., Forero-Muñoz, N.R., and Poisot, T. (2024, October 24). Connecting the dots: Bringing together scientists, policymakers, and practitioners to better implement biodiversity indicators [Side event]. GEO BON pavilion, 2024 United Nations Biodiversity Conference (COP16), Cali, Colombia. 2024

[2] Leroux, E., Mélançon, V., **Banville, F.**, Brémaud, J., Robitaille, F., and Gholamhosseini, M. (2023, March 16-17). Complexity matters: subjectivity as practice in contemporary biology [Conference]. 33rd Biology Symposium of the University of Montreal, Montreal, Qc, Canada. 2023

[1] Dansereau, G., **Banville, F.**, and Strydom, T. (2022, August 14-19). Space Oddity: Thinking about ecological networks across space [Inspire session]. ESA and CSEE Joint Meeting, Montreal, Qc, Canada. 2022

INVITED TALKS

[1] Poisot, T., **Banville, F.**, Hall, I. and Samuels, A. (2025, November 14). Rethinking One Health through biodiversity [Panel discussion]. Symposium Une seule santé, GEO BON One Health Working Group session, Brossard, Canada. 2025

CONTRIBUTED TALKS

[15] **Banville, F.**, Carlson, C., Eiffener, E., Munoz Acevedo, G., Paz Velez, A., and Poisot, T. (2026, February 24). Suivre la biodiversité pour protéger la santé humaine, animale et environnementale [Oral presentation]. 2026 QCBS Symposium, Montreal, Qc, Canada. 2026

[14] **Banville, F.** and Poisot, T. (2025, November 14). Relevance of biodiversity indicators for One Health action tracks [Oral presentation]. Symposium Une seule santé, GEO BON One Health Working Group session, Brossard, Canada. 2025

[13] **Banville, F.** and Poisot, T. (2025, October 23). Biodiversity monitoring and biosurveillance are fellow travellers [Oral presentation]. Living Data 2025, Bogota, Colombia. 2025

[12] **Banville, F.**, Carlson, C., Paz Velez, A., and Poisot, T. (2025, October 23). Monitoring biodiversity for human, animal, and environmental health [Oral presentation]. Living Data 2025, Bogota, Colombia. 2025

[11] **Banville, F.**, Carlson, C., Paz Velez, A., and Poisot, T. (2025, July 28). Monitoring biodiversity for human, animal, and environmental health [Webinar presentation]. GEO BON One Health Working Group Webinar: Towards biodiversity and health indicators, virtual. 2025

[10] **Banville, F.**, Gravel, D., and Poisot, T. (2025, July 6-9). Towards a maximum entropy theory of food webs [Conference presentation]. 2025 Annual Conference of the Canadian Society for Ecology and Evolution, Sherbrooke, Qc, Canada. 2025

[9] **Banville, F.**, Gravel, D., and Poisot, T. (2025, February 24-26). Vers une théorie de l'entropie maximale des réseaux trophiques [Conference presentation]. 2025 QCBS Symposium, Longueuil, Qc, Canada. 2025

[8] **Banville, F.**, Gravel, D., and Poisot, T. (2024, March 21-22). Quoi, quand et où manger ? À la découverte des interactions trophiques entre contraintes et incertitudes [Conference presentation]. 34th Biology Symposium of the University of Montreal, Montreal, Qc, Canada. 2024

- [7] **Banville, F.**, Gravel, D., and Poisot, T. (2024, February 19-21). Quoi, quand et où manger ? À la découverte des interactions trophiques entre contraintes et incertitudes [Conference presentation]. 2024 QCBS Symposium, Montreal, Qc, Canada. 2024
- [6] **Banville, F.**, Gravel, D., and Poisot, T. (2023, May 8-12). Comment les réseaux de prédateurs et de proies sont-ils structurés dans les milieux naturels? [Conference presentation]. 90e congrès de l'ACFAS, Montreal, Qc, Canada. 2023
- [5] **Banville, F.**, Gravel, D., and Poisot, T. (2022, August 14-19). What constrains food webs? A maximum entropy model for predicting their structure with minimal biases [Conference presentation]. 2022 Annual Meeting of the Ecological Society of America (ESA), Montreal, Qc, Canada. 2022
- [4] **Banville, F.**, Gravel, D., and Poisot, T. (2022, March 25). Food webs of maximum entropy: A story of ecology and stochasticity [Conference presentation]. 32nd Biology Symposium of the University of Montreal, Montreal, Qc, Canada. 2022
- [3] **Banville, F.**, Gravel, D., and Poisot, T. (2020, October 22). Predicting networks of species interactions [Conference presentation]. IVADO Digital October 2020, virtual. 2020
- [2] **Banville, F.**, MacDonald, A.A.M., Gravel, D., and Poisot, T. (2020, February 19). How to estimate network structure without interaction data [Conference presentation]. Extreme Climate Events Symposium 2020, Toronto, On, Canada. 2020
- [1] **Banville, F.**, MacDonald, A.A.M., Gravel, D., and Poisot, T. (2019, December 18-20). How to estimate network structure without data [Conference presentation]. 10th Annual QCBS Symposium, Montreal, Qc, Canada. 2019

LIGHTNING TALKS

- [5] **Banville, F.**, Hughes, A., Millette, K., and Poisot, T. (2026, April 6). Le Jenga de la biodiversité et de la santé [Conference short presentation]. La jeunesse interpelle le monde, One Health Summit, Lyon, France. 2026
- [4] **Banville, F.**, Gravel, D., and Poisot, T. (2022, March 29). Interactions entre espèces : une histoire d'écologie et de hasard [Conference short presentation]. My IVADO research project in 180 seconds, Montreal, Qc, Canada. 2022
- [3] **Banville, F.**, Gravel, D., and Poisot, T. (2021, October 28). Predicting food webs across space: First estimates of food-web structure derived from species richness [Conference short presentation]. IVADO Digital October 2021, virtual. 2021
- [2] **Banville, F.**, Gravel, D. and Poisot, T. (2020, December 14-16). Trophic-METE: A parsimonious theory of food-web structure [Conference short presentation]. 11th Annual QCBS Symposium, virtual. 2020
- [1] **Banville, F.**, Vissault, S., Bélisle, Z., Hoebeke, L., Stock, M., Szefer, P., and Poisot, T. (2020, July 29-31). Analyzing species interaction networks in Julia [Conference short presentation]. Juliacon 2020, virtual. 2020

POSTERS

- [3] **Banville, F.**, Carlson, C., Paz Velez, A., and Poisot, T. (2025, July 6-9). Monitoring biodiversity for human, animal, and environmental health [Poster presentation]. 2025 Annual Conference of the Canadian Society for Ecology and Evolution, Sherbrooke, Qc, Canada. 2025
- [2] **Banville, F.**, Gravel, D. and Poisot, T. (2021, December 8-10). Given limited knowledge, what can we say about a food web's properties? [Poster presentation]. 12th Annual QCBS Symposium, virtual. 2021

PHD DEFENSE

Banville, F. (2024, December 2). Vers une théorie de l'entropie maximale des réseaux trophiques [PhD defense]. University of Montreal, Montreal, Qc, Canada.

2024

Teaching and mentoring

TEACHING ASSISTANCE

F-2024	Head teaching assistant , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
F-2024	Teaching assistant , BIO2043 Statistique pratique pour sciences de la vie	<i>U. de Montréal</i>
S-2024	Head teaching assistant , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
W-2024	Head teaching assistant , BIO2042 Biostatistique 2	<i>U. de Montréal</i>
F-2023	Head teaching assistant , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
S-2023	Teaching assistant , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
W-2023	Teaching assistant , BIO2042 Biostatistique 2	<i>U. de Montréal</i>
F-2022	Head teaching assistant , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
S-2022	Teaching assistant , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
W-2022	Grader , BIO2811 Dynamique des populations	<i>U. de Montréal</i>
F-2021	Grader , BIO1001 Méthodes de recherche en biologie (TP)	<i>U. de Montréal</i>
S-2021	Teaching assistant , BIO6065 École d'été en synthèse écologique de données	<i>U. de Montréal</i>
W-2021	Grader , BIO2811 Dynamique des populations	<i>U. de Montréal</i>
F-2020	Teaching assistant , BIO3043 Théorie des réseaux	<i>U. de Montréal</i>

TUTORING AND MENTORING

F-2024	Tutor , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
S-2024	Tutor , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
W-2024	Tutor , BIO2042 Biostatistique 2	<i>U. de Montréal</i>
F-2023	Tutor , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
S-2023	Tutor , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
W-2023	Tutor , BIO2042 Biostatistique 2	<i>U. de Montréal</i>
S-2022	Tutor , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
W-2022	Mentor , IVADO's Data.Trek training program	<i>virtual</i>
S-2021	Tutor , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
S-2020	Tutor , BIO2041 Biostatistique 1	<i>U. de Montréal</i>
W-2020	Mentor , IVADO's Data.Trek training program	<i>virtual</i>

WORKSHOP FACILITATION

S-2025	Presentation assistant , CSEE workshop, Data analysis and visualization in Python for ecologists (1 day)	<i>Sherbrooke, CAN</i>
W-2025	Co-presenter , BIOS ² workshop, R and Git: From code to collaboration (1 day)	<i>Montreal, CAN</i>
S-2022	Co-presenter , Poisot lab Twitch workshop, Species distribution models in Julia (3 hours)	<i>virtual</i>
F-2021	Co-presenter , Software carpentry, Programming in R (4 days)	<i>Rimouski, CAN</i>
F-2021	Presentation assistant , Software carpentry, Introduction to Git (1 day)	<i>Rimouski, CAN</i>
W-2021	Presentation assistant , QCBS R workshops, Generalized linear models in R (3 hours)	<i>virtual</i>
W-2021	Presentation assistant , QCBS R workshops, Generalized additive models in R (3 hours)	<i>virtual</i>
W-2020	Lead presenter , IVADO's Data.Trek program, Machine learning and ecological networks in Julia (2 hours)	<i>Montreal, CAN</i>
W-2020	Co-presenter , IVADO's Data.Trek program, Introduction to R (3 hours)	<i>Montreal, CAN</i>
F-2019	Lead presenter , QCBS R workshops, Programming in R (3 hours)	<i>Montreal, CAN</i>

Internships

PHD INTERNSHIPS

Research agent (internship 2)

Group on Earth Observations Biodiversity Observation Network (GEO BON)

- forecasting beta diversity in Quebec and Colombia under different climate change scenarios
- supervision: Drs. Andrew Gonzalez (McGill University) and Timothée Poisot

Montreal, CAN

May 2021 - Aug. 2021

Event co-organizer (internship 1)

Institute for Data Valorization (IVADO)

- organizing the Data.Trek training program
- supervision: Barbara Decelle

Montreal, CAN

Jan. 2020 - Apr. 2020

BSC INTERNSHIPS - BIOLOGY

Research assistant (internship 3)

Université de Montréal, département de sciences biologiques

- analyzing the impact of environmental variables on lakes' water layers in Quebec
- supervision: Drs. Jean- François Lapierre and Roxane Maranger

Montreal, CAN

May 2019 - Aug. 2019

Research assistant (internship 2)

Université de Montréal, département de sciences biologiques

- estimating mercury fluxes in Quebec rivers
- supervision: Drs. Jean- François Lapierre and Marc Amyot

Montreal, CAN

May 2018 - Aug. 2018

Research assistant (internship 1)

Institut de recherche en biologie végétale (IRBV)

- understanding the growth of potato ovules in vitro
- supervision: Drs. Daniel Philippe Matton and Valentin Joly

Montreal, CAN

May 2017 - Aug. 2017

BSC INTERNSHIPS - MATHEMATICS

Research agent (internship 1)

Collège des médecins du Québec

- analyzing the demography of medical doctors in Quebec and the quality of their work using survey results
- supervision: Johanne Thiffault

Montreal, CAN

Jan. 2015 - July 2015

Working groups

WG participant

GEO BON One Health working group

- objective: mainstreaming environmental expertise in the One Health approach
- led by Timothée Poisot

virtual

Jan. 2025 - May 2026

WG co-organizer

Computational Biodiversity Science and Services (BIOS²)

- title: Black holes and revelations: Identifying priority sampling locations for local food webs in CAN
- led by Gabriel Dansereau, **Francis Banville**, Michael Catchen, and Tanya Strydom

Montreal, CAN

Sept. 2022

WG participant

Canadian Institute of Ecology and Evolution

- title: Assembling, predicting and refining a predator-prey metaweb for CAN
- led by Dominique Caron and Tanya Strydom

virtual

Jan. 2021 - Mar. 2022

WG participant

Living Data Project

- title: Finding indicator species by assessing the utility of sampled abundance indices
- led by Robin Freeman, Jessica Currie, and Valentina Marconi

virtual

Sept. 2021

International delegations

Delegate to the One Health Summit

Les Offices jeunesse internationaux du Québec (LOJIQ)

Lyon, France

Apr. 2026

Delegate to the COP16 on biodiversity

Group on Earth Observations Biodiversity Observation Network (GEO BON)

Cali, Colombia

Oct. 2024

Services and outreach

Peer reviewer

Scientific journals

- Human and Ecological Risk Assessment: An International Journal (1 peer review)
- Oikos journal (1 peer review)
- Journal of Open Source Software (1 peer review)

Virtual

Jan. 2025 - present

Project participant

Blitz the Gap 2025

- helped fill gaps in our knowledge of biodiversity in Canada

Saint-Hippolyte, CAN

July 2025

Volunteer

Quebec Center for Biodiversity Science (QCBS)

- provided help during the annual symposium, including the moderation of oral presentations

Longueuil, CAN

Feb. 2025

Speaker

La Nuit des chercheuses et des chercheurs, Espace pour la vie

- presented my research one on one with the general public

Montreal, CAN

Nov. 2021-23

Co-organizer of the annual symposium of the department of biological sciences

Association des étudiants-chercheurs en biologie de l'Université de Montréal (AECBUM)

- organized the theme and talks of the symposium and the caterers

U. de Montréal

Aug. 2022 - Apr. 2023

Student mentor

Centre de l'engagement étudiant

- provided assistance to new students on campus

U. de Montréal

Sept. 2022 - Dec. 2022

Environmental Coordinator

Association étudiante de biologie de l'Université de Montréal (AEBUM)

- organized awareness-raising activities related to the environment

U. de Montréal

Feb. 2016 - Aug. 2016

Treasurer

Club Végé de l'Université de Montréal

- organized awareness-raising activities related to meat consumption and managed the club's finances

U. de Montréal

Jan. 2015 - Aug. 2015

Completed graduate courses and certificates

CREDITED COURSES

W-2026 **PPA6617**, Gestion de classe en enseignement supérieur

U. de Montréal

F-2025 **PLU6035**, Enseigner au collégial aujourd'hui

U. de Montréal

S-2025 **PPA6075**, Apprentissage en enseignement supérieur

U. de Montréal

W-2025 **ETA6045**, Évaluation en enseignement supérieur

U. de Montréal

W-2021 **BIO860M**, Séminaire thématique en écologie

UQAM

F-2019 **BIO6037**, Analyse des réseaux écologiques

U. de Montréal

S-2019 **BIO6063**, Travail dirigé 1

U. de Montréal

S-2019 **BIO6065**, École d'été en synthèse écologique de données

U. de Montréal

W-2019 **BIO6032**, Biologie computationnelle et modélisation

U. de Montréal

W-2019 **BIO6033**, Méthodes quantitatives en biologie

U. de Montréal

W-2019 **BIO611**, Progrès en phylogénie systématique

U. de Montréal

W-2019 **MSO6028**, Introduction aux théories de la mesure

U. de Montréal

F-2018 **BIO6004**, Communication scientifique

U. de Montréal

F-2018 **BIO6077**, Analyse quantitative des données

U. de Montréal

F-2018 **BIO6260**, Génomique microbienne

U. de Montréal

F-2018 **GEO6321**, Travaux pratiques en géomatique

U. de Montréal

EXTRACURRICULAR COURSES

S-2023 **EFI**, Short Course on Forecasting for Decision-Making: An Epidemiological and Ecological Perspective

U. of Toronto

S-2023 **ECL807**, Advanced Field School in Computational Ecology 2023

U. de Sherbrooke

F-2022 **LDP**, Scientific collaboration in ecology and evolution

Living Data Project

F-2021 **LDP**, Synthesis statistics for ecology and evolution

Living Data Project

S-2021 **ECL807**, École d'été en modélisation de la biodiversité 2021

U. de Sherbrooke

EXTRACURRICULAR CERTIFICATES

S-2024 **CIT**, Certified Carpentries Instructor

W-2022 **LDP**, Cert. in Synthetic and Collaborative Science

The Carpentries
Living Data Project

Affiliations and professional memberships

2025-26	Postdoc member , The Viral Emergence Research Initiative (Verena)	USA
2025-26	Postdoc member , Groupe de recherche en épidémiologie des zoonoses et santé publique (GREZOSP)	U. de Montréal
2024-26	Certified Carpentries Instructor , The Carpentries	USA
2023-26	Student and postdoc member , Group on Earth Observations Biodiversity Observation Network (GEO BON)	CAN
2019-26	Student and postdoc member , Quebec Centre for Biodiversity Science (QCBS)	Quebec, CAN
2019-26	Lab member , Quantitative and Computational Ecology Lab	U. de Montréal
2025	Lab member , The Carlson Lab	Yale U.
2019-25	Fellow , Computational Biodiversity Science & Services program (BIOS ²)	CAN
2019-24	Lab member , Integrative Ecology Lab	U. de Sherbrooke
2023	Student member , Ecological Forecasting Initiative (EFI)	USA
2023	Student member , Association canadienne-française pour l'avancement des sciences (ACFAS)	Quebec, CAN
2019-23	Scholarship recipient , Institute for Data Valorization (IVADO)	Quebec, CAN
2022	Student member , Ecological Society of America (ESA)	USA